USN

[22CSDE21]

THIRD SEMESTER B.Tech EXAMINATION, DECEMBER-2023

INTRODUCTION TO ARTIFICIAL INTELLIGENCE &MACHINE LEARNING

Time: 3 Hours Maximum **Marks: 100**

Instructions: Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A	**05 X 02=10**		
1.	a.	Define artificial intelligence and list out the applications and advantages of AI.	L-1	CO1	**02**
	b.	Write short notes on feature selection	L-1	CO2	**02**
	c.	List the applications of clustering.	L-2	CO3	**02**
	d.	What are the advantages of ensemble learning?	L-1	CO4	**02**
	e.	Define deep learning	L-1	CO5	**02**
		PART-B	**06 X 05=30**		
2.	a.	Explain the life cycle/process of machine learning.	L-3	CO1	**05**
		OR			
	b.	Discuss the Linear Regression. List out the applications of Linear Regression in the real world.	L-4	CO1	**05**
3.	a.	Explain in detail about Decision tree in machine learning.	L-3	CO2	**05**
		OR			
	b.	Describe about evaluation measures for classification and regression in supervised learning models.	L-2	CO2	**05**
4.	a.	Explain types of machine learning models.	L-3	CO3	**05**
		OR			
	b.	List and explain the Advantages and Disadvantages of K-Means Clustering.	L-2	CO3	**05**
5.	a.	How to verify the condition for consistency of a system of linear equation? Explain	L-4	CO4	**05**
		OR			
	b.	How do you solve eigenvalues and eigenvectors of a square matrix manually? Explain	L-4	CO4	**05**
6.	a.	Define neural network and list out the applications of neural network	L-3	CO5	**05**
		OR			
	b.	List and explain the applications of deep learning.	L-2	CO5	**05**
7.	a.	What is confusion matrix and write down the 2X2 confusion matrix for evaluation of classification.	L-1	CO2	**05**

		OR			
	b.	Write short notes on Learning from Unclassified Data	L-1	CO3	**05**
		PART-C	**06X10=60**		
8.	**a.**	Explain the missing data imputation using mean, median and mode statistical measures.	L-3	CO1	**10**
		OR			
	b.	Linear regression problems : Find linear regression for the following data. Also estimate the value of y at x=10. X 2 4 6 7 8 y 4 8 10 12 14	L-4	CO1	**10**
9.	**a.**	With example, explain the concept of decision tree and how to evaluate the decision tree for classification	L-3	CO2	**10**
		OR			
	b.	Discuss the over fitting and under fitting.	L-3	CO2	**10**
10.	**a.**	Explain the Hierarchical Agglomerative Clustering	L-3	CO3	**10**
		OR			
	b.	**i.** Difference between clustering and classification	L-4	CO3	**05**
		ii. Explain soft and hard clustering with examples	L-3	CO3	**05**
11.	**a.**	Explain in detail about ensemble machine learning technique with examples	L-3	CO4	**10**
		OR			
	b.	Define rank of a matrix. Find the rank of the following matrix by reducing it to the row echelon form. $A = \begin{pmatrix} 0 & 1 & 2 & 1 \\ 1 & 2 & 3 & 2 \\ 3 & 1 & 1 & 3 \end{pmatrix}$	L-2	CO4	**10**
12.	**a.**	Explain in detail about CNN architecture with diagram.	L-3	CO5	**10**
		OR			
	b.	Discuss about architecture of RNN	L-3	CO5	**10**
13.	**a.**	Discuss about the distribution of dataset into train and test data.	L-3	CO5	**10**
		OR			
	b.	**i.** Write short notes on decision boundary.	L-1	CO5	**05**
		ii. Explain the Neural network algorithm.	L-3	CO5	**05**